No Code Research Experiments

Using PsychoPy

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Outline

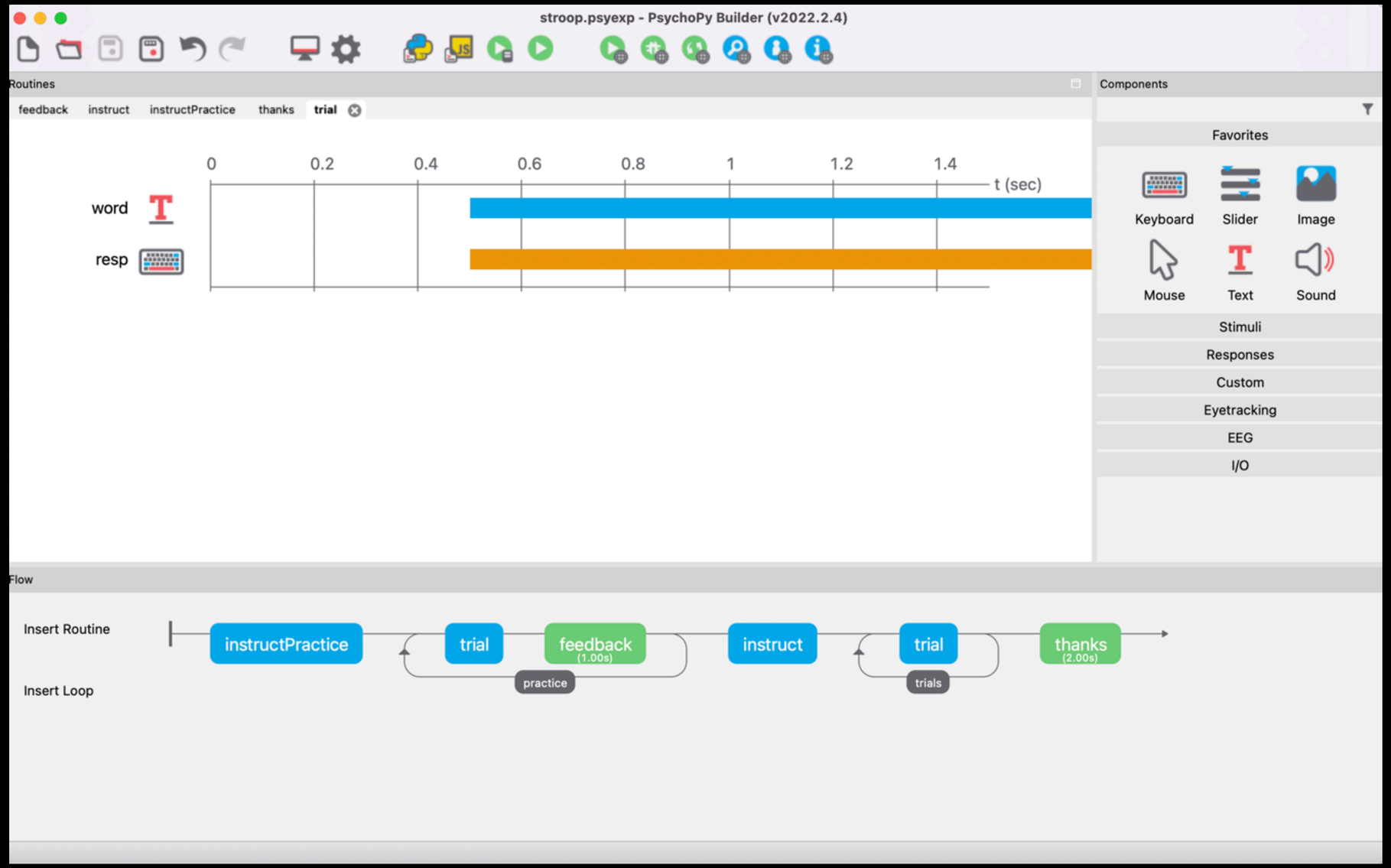
- What kind of data can we collect?
- What is PscyhoPy?
- Example experiment
- How to build experiments

PsychoPy

- Designed for psychology experiments
- Extremely repeatable
  - Can calibrate letter size, screen color and brightness, volume, frame rate
- Easy to use with no programming

PsychoPy Main Ideas

- Flow - the whole experiment, end to end
- Routines - Each part of the experiment, think screens or questions
- Components - Inputs and outputs used to build a routine



Say the colors of the words

blue

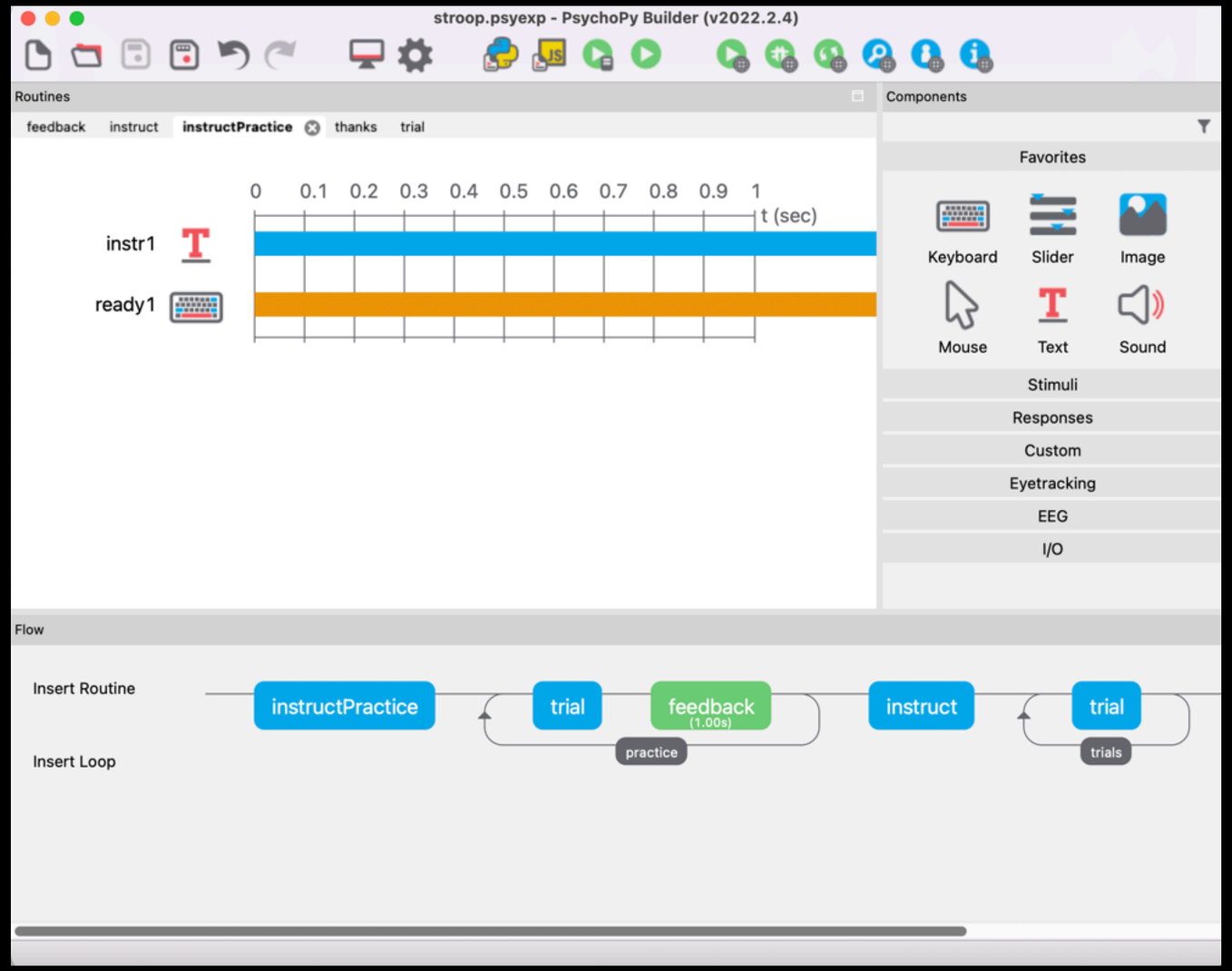
red

blue

An Example Project

The Stroop Effect

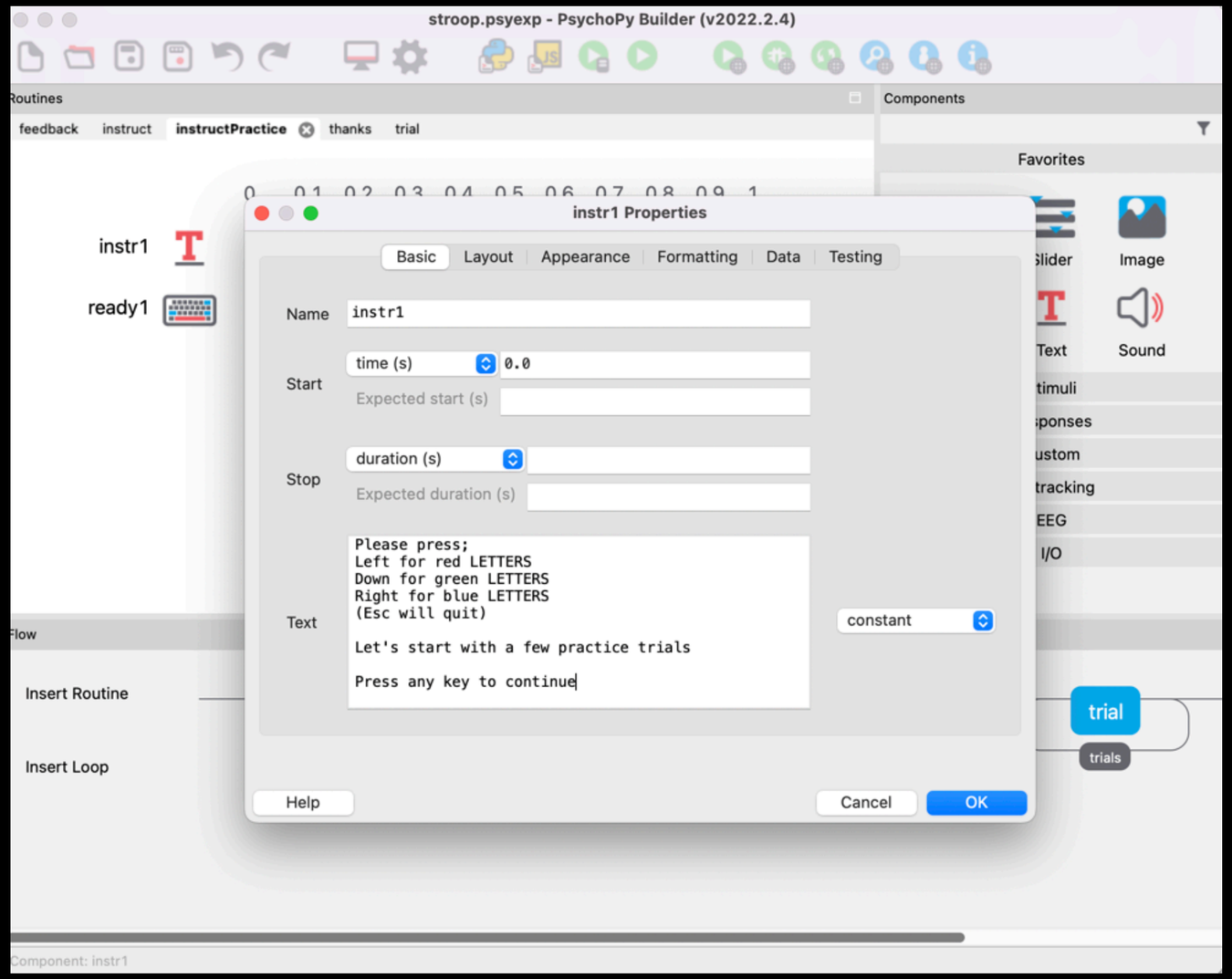
- The "instructions" routine
- The bars represent time
- Clicking on components shows its options
- Constant means the setting never changes



An Example Project

The Stroop Effect

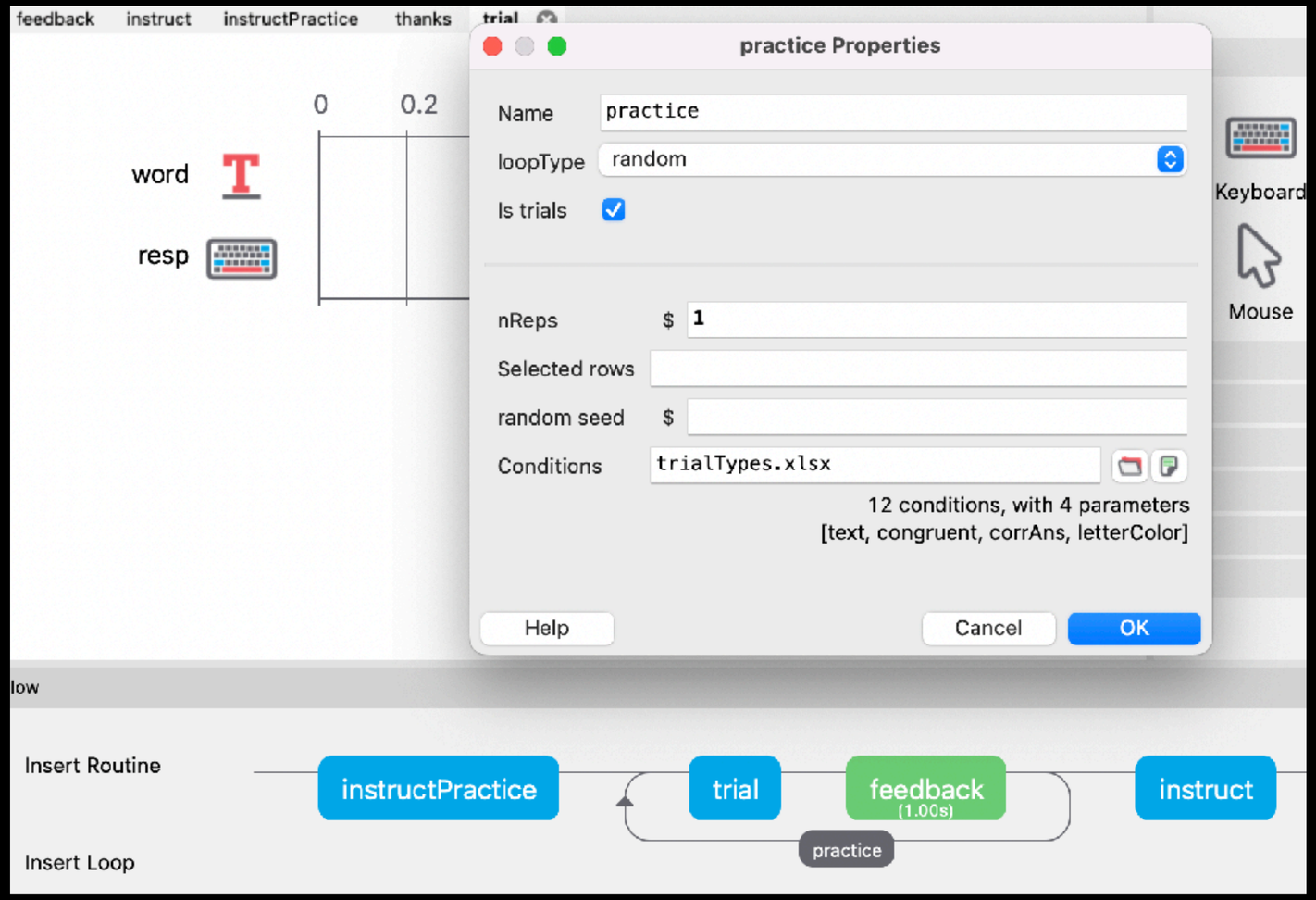
- Everything is customisable
- Can stay the same or change everytime it is shown



Loops

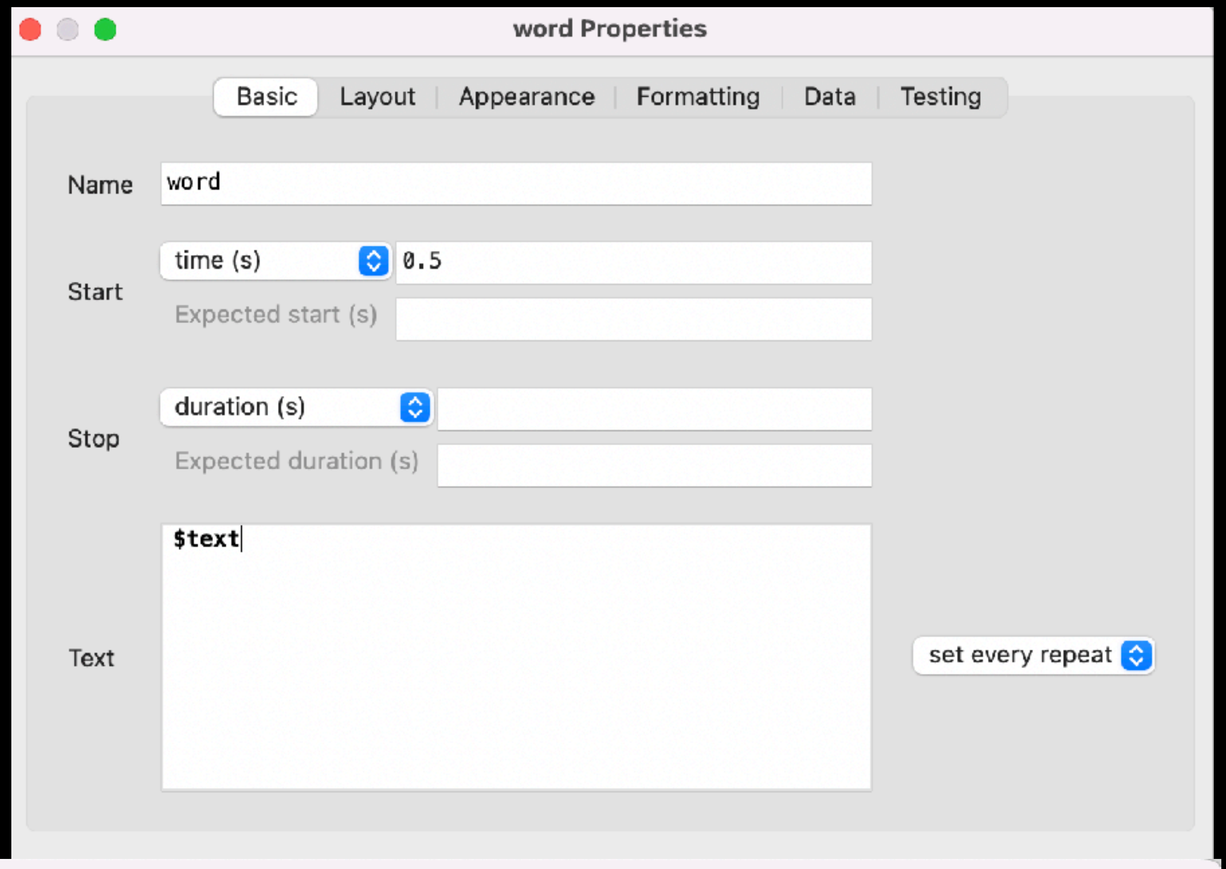
Repeating Sections of the Experiment

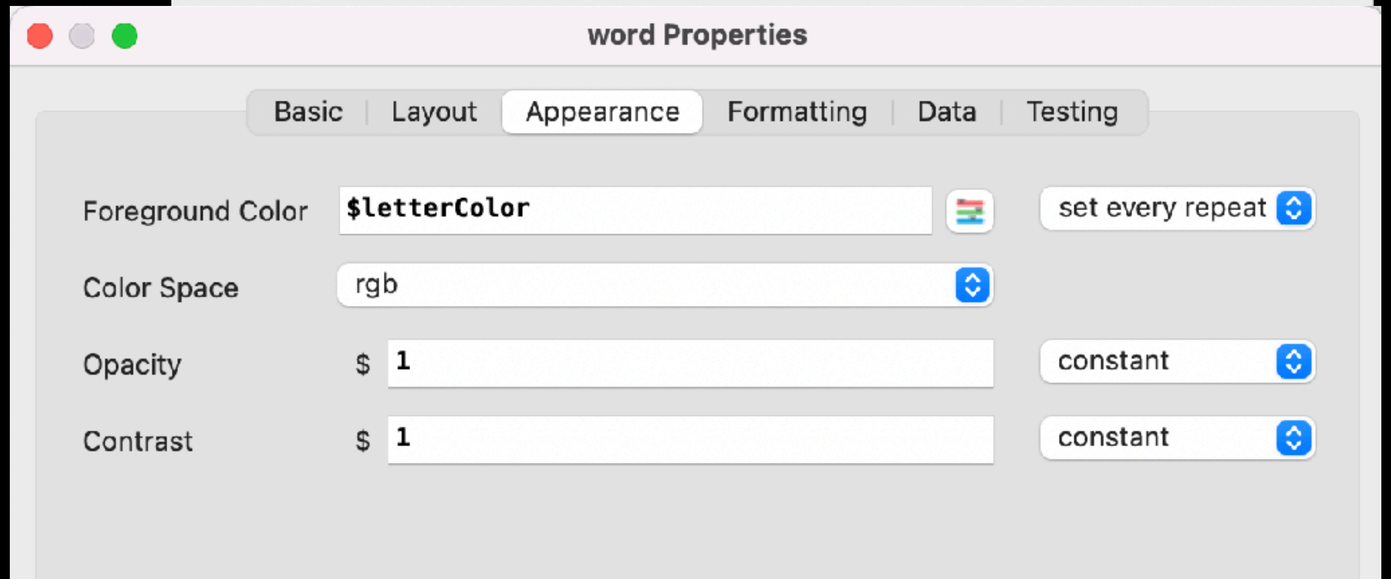
- Each time round the loop is one line of a spreadsheet
- nReps (how many times to repeat the whole spreadsheet)
- loopType (what order to show the rows)
  - Random - random order with each repeat shuffled seperately
  - Sequential - same order as spreadsheet
  - Full random - multiple repeats mixed together in random order



Trials

- Anything starting with a $ symbol is data from your spreadsheet
- It must match the title of the spreadsheet column **exactly**

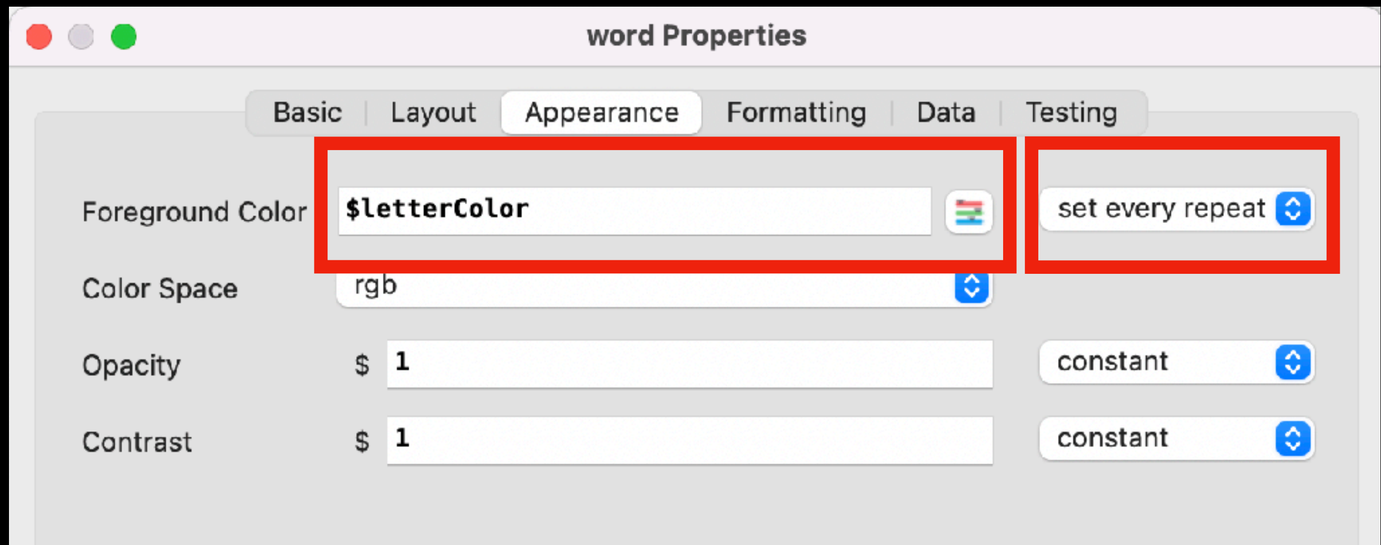




Trials

- Column title matches exactly
- The loop is random, so each repeat is a random color
- Don't forget to choose "set every repeat"

text	letterColor	corrAns	congruent	Size
red	red	eft	cong	0.2
red	red	eft	cong	0.4
red	green	down	incong	0.2
red	blue	ight	incong	0.2
green	red	eft	incong	0.6
green	green	down	cong	0.2
green	green	down	cong	0.1
green	blue	ight	incong	0.2
blue	red	eft	incong	0.3
blue	green	down	incong	0.2
blue	blue	ight	cong	0.2
blue	blue	ight	cong	0.2



Collected Data

- Data is automatically saved in a data folder
- One spreadsheet per session

text	letterC	corrAns	congru	n	resp.keys_	resp.rt_	resp.r	resp.rt_	order
red	red	left	cong	1	'left'	0.717	0.717	0	5
red	red	left	cong	1	'left'	0.765	0.765	0	9
red	green	down	incong	1	'down'	0.682	0.682	0	6
red	blue	right	incong	1	'right'	1.065	1.065	0	7
green	red	left	incong	1	'left'	0.946	0.946	0	8
green	green	down	cong	1	'left'	0.684	0.684	0	2
green	green	down	cong	1	'down'	0.986	0.986	0	11
green	blue	right	incong	1	'right'	0.530	0.530	0	1
blue	red	left	incong	1	'left'	0.984	0.984	0	3
blue	green	down	incong	1	'down'	0.607	0.607	0	10
blue	blue	right	cong	1	'right'	1.575	1.575	0	0
blue	blue	right	cong	1	'right'	0.696	0.696	0	4
extraInfo									
particip	Laurie								
session	2								
date	2022-10-28_15h07.19.000								
expNar	stroop								
psycho	2022.2.4								
frameRate									

Collected Data

- The original trial data is included

text	letterC	corrAns	congru	n	resp.keys_	resp.rt_	resp.r	resp.rt_	order
red	red	left	cong	1	'left'	0.717	0.717	0	5
red	red	left	cong	1	'left'	0.765	0.765	0	9
red	green	down	incong	1	'down'	0.682	0.682	0	6
red	blue	right	incong	1	'right'	1.065	1.065	0	7
green	red	left	incong	1	'left'	0.946	0.946	0	8
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extraInfo									
particip	Laurie								
session	2								
date	2022-10-28_15h07.19.000								
expNar	stroop								
psycho	2022.2.4								
frameRate									

Collected Data

- Responses are included
- Time in seconds is included
  - Very precise numbers, but how accurate? (One frame ~16ms)

text	letterC	corrAns	congru	n	resp.keys_	resp.rt_	resp.r	resp.rt_	order
red	red	left	cong	.	'left'	0.717	0.717	0	5
red	red	left	cong	.	'left'	0.765	0.765	0	9
red	green	down	incong	.	'down'	0.682	0.682	0	6
red	blue	right	incong	.	'right'	1.065	1.065	0	7
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extraInfo									
particip	Laurie								
session	2								
date	2022-10-28_15h07.19.000								
expNar	stroop								
psycho	2022.2.4								
frameRate									

Collected Data

- The shuffled order is recorded

text	letterC	corrAns	congru	n	resp.keys_	resp.rt_	resp.r	resp.r	_order
red	red	left	cong	1	'left'	0.717	0.717	0	5
red	red	left	cong	1	'left'	0.765	0.765	0	9
red	green	down	incong	1	'down'	0.682	0.682	0	6
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extraInfo									
particip	Laurie								
session	2								
date	2022-10-28_15h07.19.000								
expNar	stroop								
psycho	2022.2.4								
frameRate									

Collected Data

- Extra data
- Useful for repeated sessions over weeks and months with the same people

text	letterC	corrAns	congru	n	resp.keys_	resp.rt_	resp.r	resp.rt_	order
red	red	left	cong	1	'left'	0.717	0.717	0	5
red	red	left	cong	1	'left'	0.765	0.765	0	9
red	green	down	incong	1	'down'	0.682	0.682	0	6
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blue	blue	right	cong	1	'right'	0.696	0.696	0	4
extraInfo									
particip	Laurie								
session	2								
date	2022-10-28_15h07.19.000								
expNar	stroop								
psycho	2022.2.4								
frameRate									

Building from Scratch

Planning

- What data are you trying to collect?
- What stimuli will you use? (And how will you create them?)
- Do you need any extra equipment for your experiment?
- Will people do you experiment once or many times?

Creating our trials

- Minimal Pairs

collect	correct
glamour	grammar
glass	grass
lace	race
lane	rain
law	raw
lead	read
leader	Reader

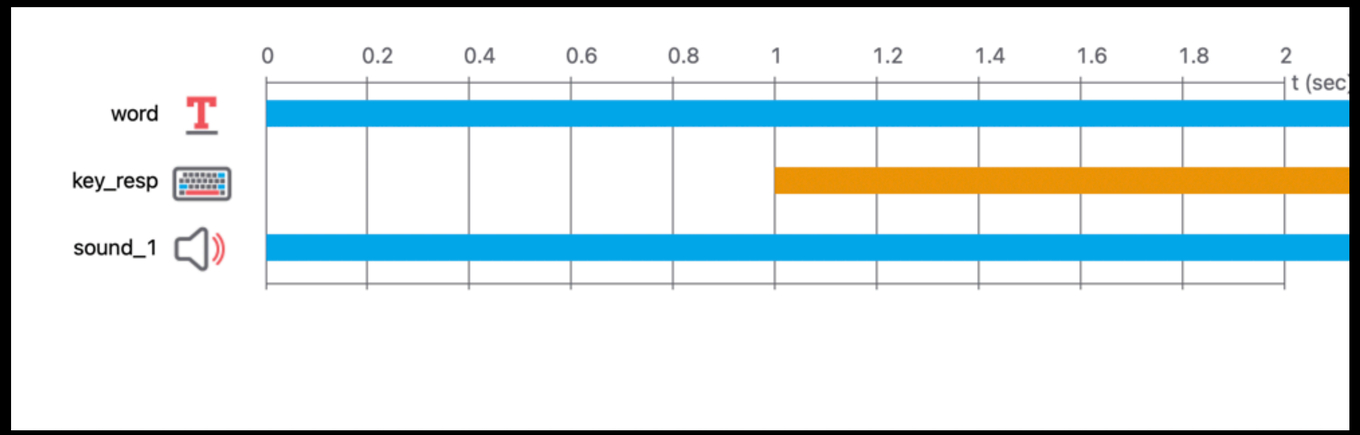
Creating our trials

- Create extra data
- Need a row for each answer
- Trial number can be helpful for finding problems
- Don't have to use them all

trial_no	l_word	r_word	correct_ans	audio_file
0	collect	correct	collect	collect.ogg
1	glamour	grammar	glamour	glamour.ogg
2	glass	grass	glass	glass.ogg
3	lace	race	lace	lace.ogg
4	lane	rain	lane	lane.ogg
5	law	raw	law	law.ogg
6	lead	read	lead	lead.ogg
7	leader	reader	leader	leader.ogg
8	collect	correct	correct	correct.ogg
9	glamour	grammar	grammar	grammar.ogg
10	glass	grass	grass	grass.ogg
11	lace	race	race	race.ogg
12	lane	rain	rain	rain.ogg
13	law	raw	raw	raw.ogg
14	lead	read	read	read.ogg
15	leader	reader	reader	reader.ogg

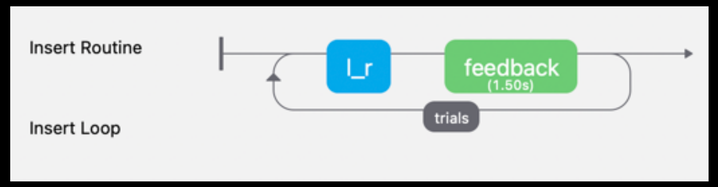
Create Routines

- Generally need some kind of stimulus and response
- Show word, play sound, get key press



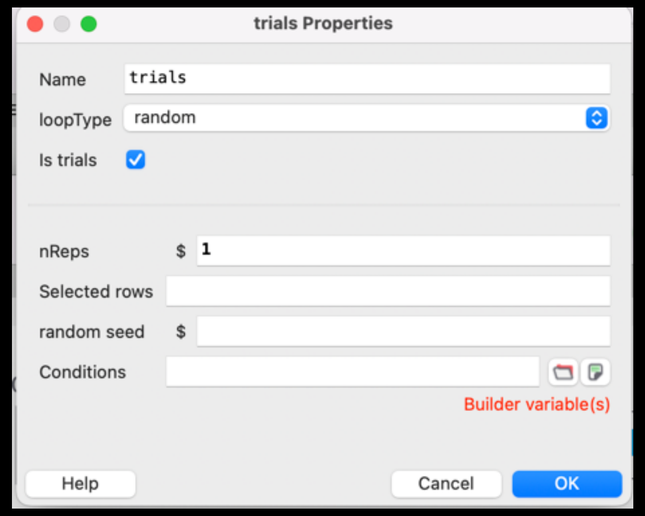
Make Flow from Routines

- Can delete and add the same routine (no drag and drop!
- Need to add loops to get add repeats and data



Common Issues/Tips

- "Builder variable" error
  - Happens if you choose a reserved word as a column name. ("filename" is a reserved word)
- Check your output data thoroughly before running your experiment with participants!



What's next?

- Hook up more hardware
  - Game controllers, cameras
  - eye-tracking, brain wave EEG
- Custom code (use python or Javascript)
- Deploy experiments online
  - pavlovia.org

Questions? Ideas?

Code, slides and contact details

- Email: laurieinjapan@gmail.com
- Twitter: @laurie_ljg
- Slides: https://lauriegriffiths.github.io/files/ResearchExperiments.pdf
- psychopy.org/